Nevertheless, this transition can be useful for other types of graphs that aren't symmetrical, so we'll leave it in (while we are using complete graphs for our examples, the algorithm can and will be applied to any graph: we'll see in the next section some examples where swapping vertices becomes crucial to getting a good result).

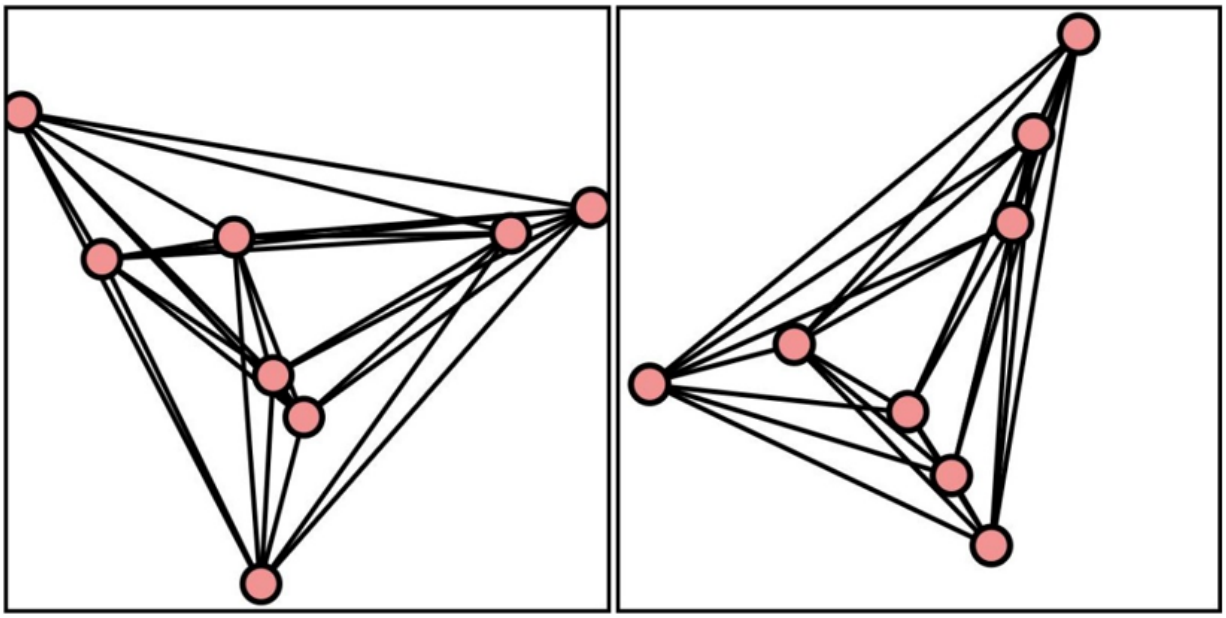


Figure 17.14 Two embeddings of complete graph K_8: on the left, one computed by the random sampling algorithm; on the right, the result of simulated annealing. Both algorithms were run for 500 iterations: random sampling could only get as low as 27 intersections; simulated annealing produced an optimal embedding with 19 crossing points.

Yet, we still need to do something different to improve our algorithm: what about choosing a single vertex and moving it randomly anywhere in the drawing area?